EXERCÍCIOS:

Calcular os Determinantes:

1) $\begin{vmatrix}1&2&1&2\\2&1&2&1\\-1&1&-2&2\\1&-1&2&-2\end{vmatrix}$ 2) $\begin{vmatrix}4&4&2&2\\-3&0&4&2\\6&2&2&0\\-2&2&6&4\end{vmatrix}$ 3) $\begin{vmatrix}4&2&2&2\\2&1&2&3\\2&2&3&4\\2&3&4&5\end{vmatrix}$

①

1) $\begin{vmatrix}1&2&1&2\\2&1&2&1\\-1&1&-2&2\\1&-1&2&-2\end{vmatrix}$

$A = \begin{bmatrix}1&2&1&2\\2&1&2&1\\-1&1&-2&2\\1&-1&2&-2\end{bmatrix}$

$A_{11} = \begin{bmatrix}1&2&1\\1&-2&2\\-1&2&-2\end{bmatrix} = 1\times1\begin{bmatrix}-2&2\\2&-2\end{bmatrix} = 1\times(-2\times(-2)-2\times2)$

$= 1\times(4-4)$

$= 1\times0 = 0$

$= 1\times(-1)\begin{bmatrix}2&1\\2&-2\end{bmatrix} = -1\times(2\times(-2)-2\times1)$

$-1\times(-4-2)$

$-1\times(-6) = 6$

$= -1\times1\begin{bmatrix}2&1\\-2&2\end{bmatrix} = -1\times(2\times2-(-2\times1))$

$= -1\times(4-(-2))$

$= -1\times6$

$= -6$

$0+6-6 = \underline{0}$

$A_{12} = \begin{bmatrix}2&2&1\\-1&-2&2\\1&2&-2\end{bmatrix} = 1\times2\begin{bmatrix}-2&2\\2&-2\end{bmatrix} = 2\times(-2\times-2-2\times2)$

$= 2\times(4-4)$

$= 2\times0$

$= 0$

$= -1\times-1\begin{bmatrix}2&1\\2&-2\end{bmatrix}$

$= 1\times(2\times(-2)-2\times1)$

$= 1\times(-4-2)$

$= 1\times(-6) = -6$

$\begin{bmatrix}2&1\\-2&2\end{bmatrix} = 1\times1(2\times2-(-2\times1))$

$= 1(4-(-2))$

$= 6$

$0-6+6$

$= \underline{0}$

$A_{13} = \begin{bmatrix}2&1&1\\-1&1&2\\1&-1&-2\end{bmatrix} = 2\times1\begin{bmatrix}1&2\\-1&-2\end{bmatrix} = 2(1\times(-2)-(-1)\times2)$

$2(-2-(-2))$

$2(0) = \underline{0}$

-1×-1 $\begin{bmatrix}1&1\\-1&-2\end{bmatrix} = 1(1\times(-2)-(-1\times1))$

$-2-(-1) = \underline{-1}$

$1\times1\begin{bmatrix}1&1\\1&2\end{bmatrix} = 1(1\times2-1\times1)$

$1(2-1) = \underline{1}$

$0-1+1 = \underline{0}$

R: $0+0+0 = 0$

②

2) $\begin{vmatrix}4&4&2&2\\-3&0&4&2\\6&2&2&0\\-2&2&6&4\end{vmatrix}$

$\begin{bmatrix}4&4&2&2\\-3&0&4&2\\6&2&2&0\\-2&2&6&4\end{bmatrix}$

$21\ \begin{bmatrix}4&2&2\\2&2&0\\2&6&4\end{bmatrix}\begin{matrix}4&2\\2&2\\2&6\end{matrix}$

$(4\times2\times4+2\times0\times2+2\times2\times6) = 56$

$-(2\times2\times2+6\times0\times4+4\times2\times2) = 24$ $\}32$

$22 = 0$

$23 = \begin{bmatrix}4&4&2\\6&2&0\\-2&2&4\end{bmatrix}\begin{matrix}4&4\\6&2\\-2&2\end{matrix}$

$(4\times2\times4+4\times0\times(-2)+2\times6\times2) = 56$

$-(-2\times2\times2+2\times0\times4+4\times6\times4)$ $\}+32$

$= \underline{88}$

$24\ \begin{bmatrix}4&4&2\\6&2&2\\-2&2&6\end{bmatrix}\begin{matrix}4&4\\6&2\\-2&2\end{matrix}$

$(\overset{48}{4\times2\times6}+\overset{(-16)}{4\times2\times(-2)}+\overset{24}{2\times6\times2})$

$-(\overset{-8}{-2\times2\times2}+\overset{16}{2\times2\times4}+\overset{144}{6\times6\times4})$

$= 56 - 152 = \underline{-96}$

$-3\times -32+1\times32+2\times(-96)$

$= 32$

③

3) $\begin{vmatrix}4&2&2&2\\2&1&2&3\\2&2&3&4\\2&3&4&5\end{vmatrix}$

$\begin{bmatrix}4&2&2&2\\2&1&2&3\\2&2&3&4\\2&3&4&5\end{bmatrix}$

$0+0+0+0 = 0$

$11 = \begin{bmatrix}1&2&3\\2&3&4\\3&4&5\end{bmatrix} = 1\times1\begin{bmatrix}3&4\\4&5\end{bmatrix} = 1(3\times5-4\times4)$

$= 15-16$

$= \cancel{1}-1$

$= -1\times2\begin{bmatrix}2&3\\4&5\end{bmatrix} = -2(2\times5-4\times3)$

$2(10-12)$

$-2\times(-2) = 4$

$= 1\times3\begin{bmatrix}2&3\\3&4\end{bmatrix} = 3(2\times4-3\times3)$

$= 3(8-9)$

$= 3\times(-1) = \underline{-3}$

$-1+4-3 =$

$= -1+1 = 0$

$12\ \begin{bmatrix}2&2&3\\2&3&4\\2&4&5\end{bmatrix} = 1\times2\begin{bmatrix}3&4\\4&5\end{bmatrix} = 2(3\times5-4\times4)$

$2(15-16)$

$2\times(-1) = -2$

$= -1\times2\begin{bmatrix}2&3\\4&5\end{bmatrix} = -2(2\times5-4\times3)$

$-2(10-12)$

$-2\times(-2) = 4$

$= 1\times2\begin{bmatrix}2&3\\3&4\end{bmatrix} = 2(2\times4-3\times3)$

$2(8-9)$

$2\times(-1) = -2$

$-2+4-2 = 0$

$13\ \begin{bmatrix}2&1&3\\2&2&4\\2&3&5\end{bmatrix} = 1\times2\begin{bmatrix}2&4\\3&5\end{bmatrix} = 2(2\times5-3\times4)$

$2(10-12)$

$2\times(-2) = -4$

$= -1\times2\begin{bmatrix}1&3\\3&5\end{bmatrix} = -2(1\times5-3\times3)$

$-2(5-9)$

$-2\times(-4) = 8$

$= 1\times2\begin{bmatrix}1&3\\2&4\end{bmatrix} = 2(1\times4-2\times3)$

$2(4-6)$

$2\times(-2) = -4$

$-4+8-4 = \underline{0}$

$14\ \begin{bmatrix}2&1&2\\2&2&3\\2&3&4\end{bmatrix} = 1\times2\begin{bmatrix}2&3\\3&4\end{bmatrix} = 2(2\times4-3\times3)$

$= 2(8-9)$

$2\times(-1) = -2$

$= -1\times2\begin{bmatrix}1&2\\3&4\end{bmatrix} = -2(1\times4-3\times2)$

$-2(4-6)$

$-2\times(-2) = 4$

$= 1\times2\begin{bmatrix}1&2\\2&3\end{bmatrix} = 2(1\times3-2\times2)$

$2(3-4) = 2\times(-1) = -2$

$-2+4-2 = 0$